Endoscopic Management of Acute and Chronic Pancreatitis

Fernando Calmet

10/24/2026

Acute Pancreatitis

Acute pancreatitis

Definition

- Acute inflammatory process of the pancreas with variable involvement of other regional tissues or remote organ systems.

Boxhoorn et al. Lancet. 2020;396(10252):726–734.
Forsmark et al. N Engl J Med. 2016;375(20):1972–1981.

Acute pancreatitis

Definition

- Acute inflammatory process of the pancreas with variable involvement of other regional tissues or remote organ systems.

Etiology

- Biliary: 40–45%
- Alcohol: 20–30%
- Idiopathic: 10–30%
- Others: hypercalcemia, hypertriglyceridemia, autoimmune, post-ERCP

Boxhoorn et al. Lancet. 2020;396(10252):726–734.
Forsmark et al. N Engl J Med. 2016;375(20):1972–1981.

Acute pancreatitis

Diagnosis (Atlanta criteria) — *2 of the following:*

- Abdominal pain consistent with acute pancreatitis
- Serum lipase (or amylase) $\geq$3 × upper limit of normal
- Characteristic findings on CT, MRI, or US

Banks et al. Gut. 2013;62(1):102–11.
Tenner et al. Am J Gastroenterol. 2024;119(3):419–437.
IAP/APA/EPC/IPC/JPS Working Group. Pancreatology. 2025;25(6):770–814.

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Management

- Moderate IV fluids
- Nutrition
- Pain control

Banks et al. Gut. 2013;62(1):102–11.
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IAP/APA/EPC/IPC/JPS Working Group. Pancreatology. 2025;25(6):770–814.

Acute pancreatitis

Risk of recurrence and chronic pancreatitis

- After index episode 10–30% of patients develop recurrent acute pancreatitis (RAP)
- 36% of patients with RAP progress to chronic pancreatitis

Yadav et al. Am J Gastroenterol. 2012;107(7):1096–103.
Sankaran et al. Gastroenterology. 2015;149(6):1490–1500.e1.
Strand et al. Gastroenterology. 2022;163(4):1107–1114.

ERCP in acute biliary pancreatitis

Acute biliary pancreatitis

- No benefit from routine ERCP regarding mortality, local or systemic complications
- ERCP recommended in patients with:
  - Acute cholangitis (within 24–72 hours)
  - Obstructive choledocholithiasis (within 72 hours)

Coutinho et al. Pancreas. 2018;47(4):444–453.
Tse et al. Cochrane Database Syst Rev. 2012;2012(5):CD009779.
IAP/APA/EPC/IPC/JPS Working Group. Pancreatology. 2025;25(6):770–814.

Approach to suspected choledocholithiasis

High likelihood $\rightarrow$ ERCP

- CBD stone on cross-sectional imaging
- Bilirubin $>$4 mg/dL *and* CBD dilation ($>$6 mm with gallbladder in situ)
- Ascending cholangitis

Intermediate likelihood $\rightarrow$ EUS, MRCP, or IOC

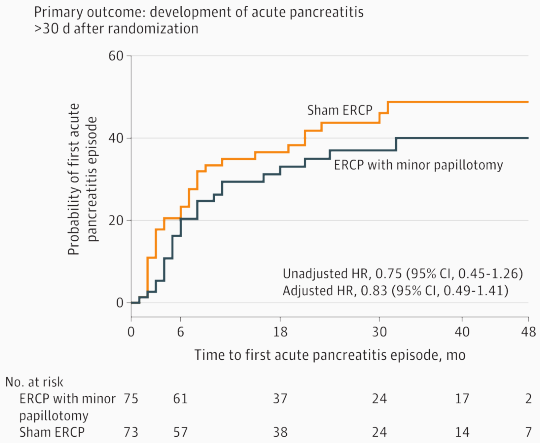
- Abnormal liver tests
- Age $>$55 years
- CBD dilation

Buxbaum et al. Gastrointest Endosc. 2019;89(6):1075–1105.e15.

ERCP in recurrent acute pancreatitis (RAP)

RAP with pancreas divisum

- 148 patients randomized to ERCP with minor papillotomy vs. sham ERCP
- Median follow-up: 34 months
- ERCP with minor papillotomy did not reduce risk of recurrent pancreatitis



Cote et al. JAMA. 2026;335(8):682–692.

Wehrmann. Endoscopy. 2011;43(3):202–7.

Das et al. Pancreatology. 2016;16(5):770–7.

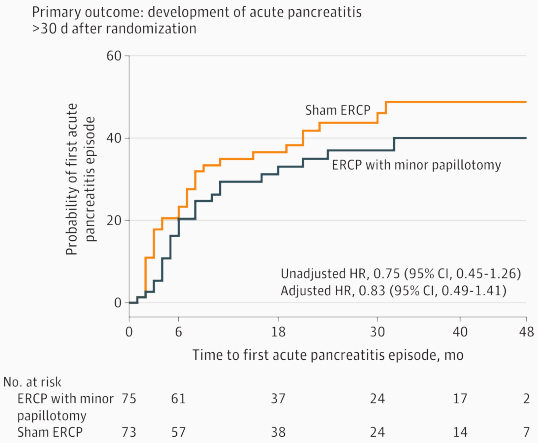
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- 148 patients randomized to ERCP with minor papillotomy vs. sham ERCP
- Median follow-up: 34 months
- ERCP with minor papillotomy did not reduce risk of recurrent pancreatitis

RAP without pancreas divisum

- Only low-quality observational studies available with mixed results



Cote et al. JAMA. 2026;335(8):682–692.

Wehrmann. Endoscopy. 2011;43(3):202–7.

Das et al. Pancreatology. 2016;16(5):770–7.

EUS in idiopathic acute pancreatitis (IAP) and idiopathic RAP (IRAP)

Role of endoscopic ultrasound (EUS)

- EUS may uncover a potential etiology in 29–88% of IAP
- Most patients initially classified as IAP found to have a biliary etiology
- Up to 5% of IAP and 12% of IRAP found to have underlying malignancy

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Strand et al. Gastroenterology. 2022;163(4):1107–1114.
Hallensleben et al. United European Gastroenterol J. 2020;8(3):340–350.

EUS in IAP and IRAP

Etiology	EUS	MRCP	*p* value
Overall	57.8%	28.9%	<0.01
Biliary	30.8%	8.1%	<0.0001
Chronic pancreatitis	9.1%	3.4%	0.0955
Pancreas divisum	3.8%	7.5%	0.1078
Malignancy	4.4%	2.0%	0.2896
IPMN	4.1%	4.6%	0.7861

IPMN, intraductal papillary mucinous neoplasm.

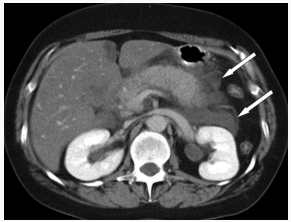
Udaikumar et al. Dig Dis Sci. 2026;71(3):1108–1118.

Complications of Acute Pancreatitis

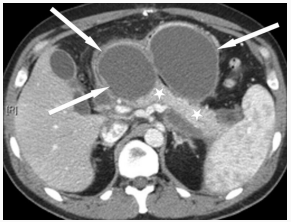
Complications of acute pancreatitis—Pancreatic fluid collections (PFCs)

Interstitial pancreatitis

Acute pancreatic fluid collection

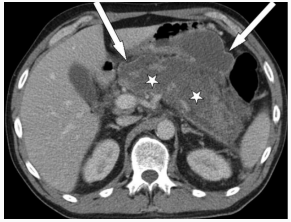


Pseudocyst



Necrotizing pancreatitis

Acute necrotic collection



Walled-off necrosis



Banks et al. Gut. 2013;62(1):102–11.

Pseudocyst drainage

- Drainage indicated if symptomatic

Sheth et al. Gastrointest Endosc. 2024;100(4):584–594.
Farias et al. Medicine (Baltimore). 2019;98(8):e14255.
Varadarajulu et al. Gastroenterology. 2013;145(3):583–90.e1.
Chhabra et al. Dig Dis Sci. 2024;69(2):335–348.

Pseudocyst drainage

- Drainage indicated if symptomatic
- EUS-guided vs. surgical drainage
  - No difference in treatment success, adverse events, and recurrence
  - EUS-guided drainage associated with shorter length of stay and lower treatment costs

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Pseudocyst drainage

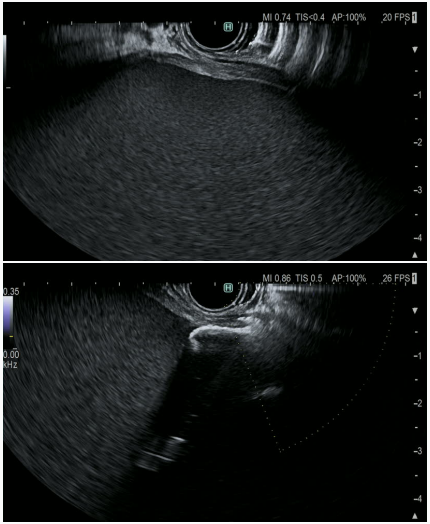
- Drainage indicated if symptomatic
- EUS-guided vs. surgical drainage
  - No difference in treatment success, adverse events, and recurrence
  - EUS-guided drainage associated with shorter length of stay and lower treatment costs
- EUS-guided vs. percutaneous drainage
  - EUS-guided drainage associated with improved treatment success, shorter length of stay, and lower need for reintervention
  - Percutaneous drainage associated with external fistulas in 20–44%

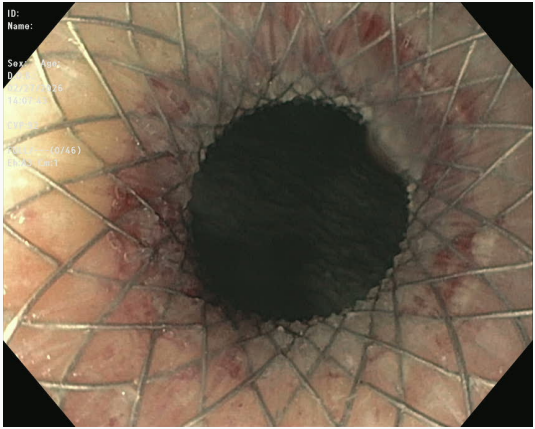
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Varadarajulu et al. Gastroenterology. 2013;145(3):583–90.e1.
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Pseudocyst—40 y.o. man with alcohol-related necrotizing pancreatitis



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Pseudocyst—40 y.o. man with alcohol-related necrotizing pancreatitis



Walled-off necrosis (WON)

Pivotal randomized controlled trials

- PANTER: upfront surgical necrosectomy vs. step-up approach
  - New-onset multiorgan failure less frequent in step-up group (12% vs. 40%, $p = 0.002$)

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van Santvoort et al. N Engl J Med. 2010;362(16):1491–502.
van Brunschot et al. Lancet. 2018;391(10115):51–58.
Bang et al. Gastroenterology. 2019;156(4):1027–1040.e3.

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- TENSION: endoscopic step-up approach vs. surgical step-up approach
  - Less pancreatic fistulas and shorter length of stay in endoscopic group

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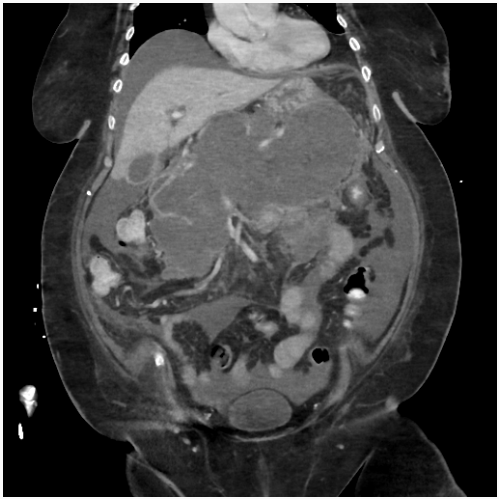
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- TENSION: endoscopic step-up approach vs. surgical step-up approach
  - Less pancreatic fistulas and shorter length of stay in endoscopic group
- MISER: endoscopic step-up approach vs. surgical step-up approach
  - Less pancreatic fistulas (0 vs. 28%, p 0.001) and lower costs in endoscopic group

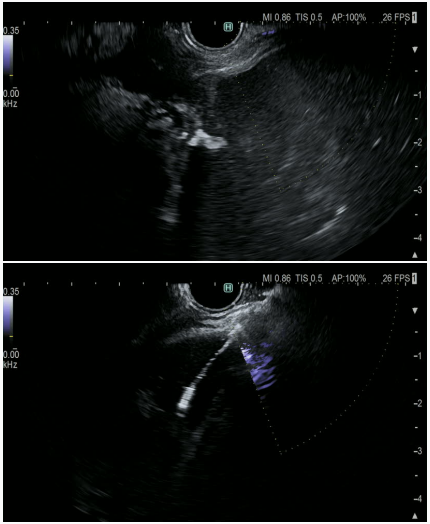
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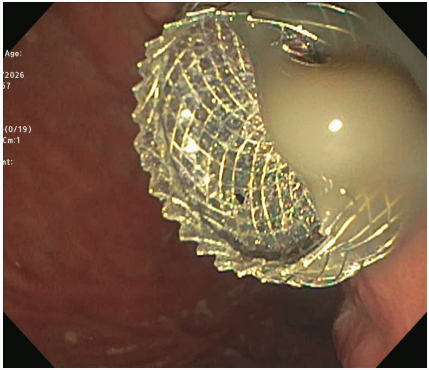
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WON—67 y.o. woman with tirzepatide-induced necrotizing pancreatitis

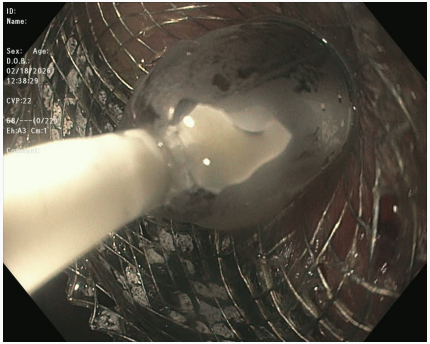


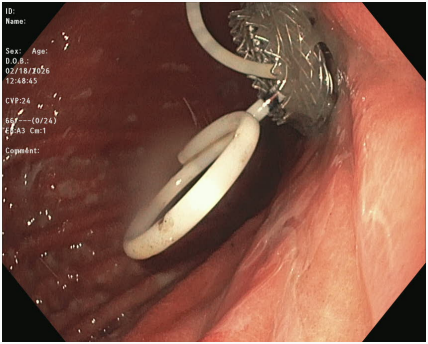
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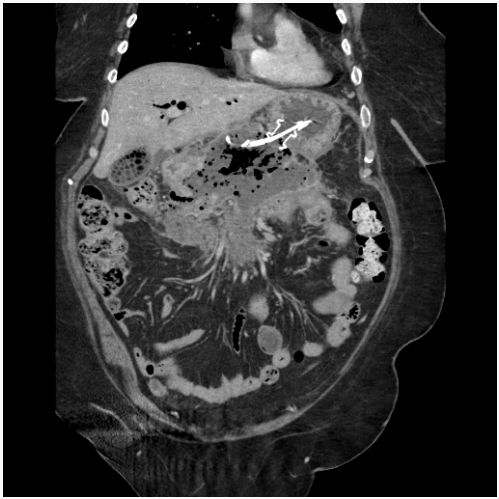


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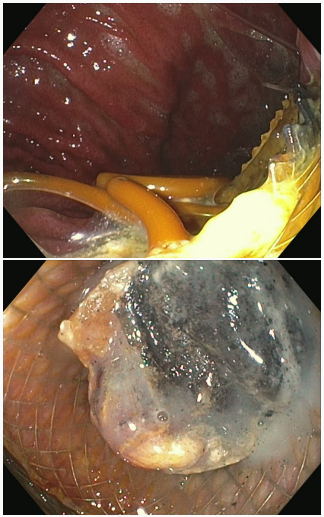


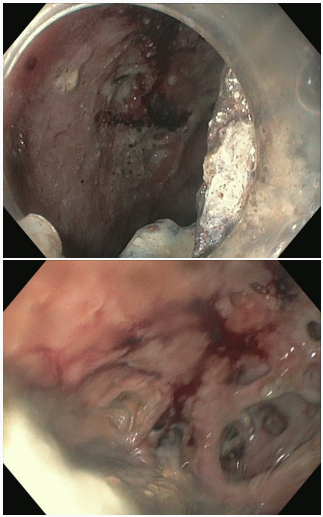


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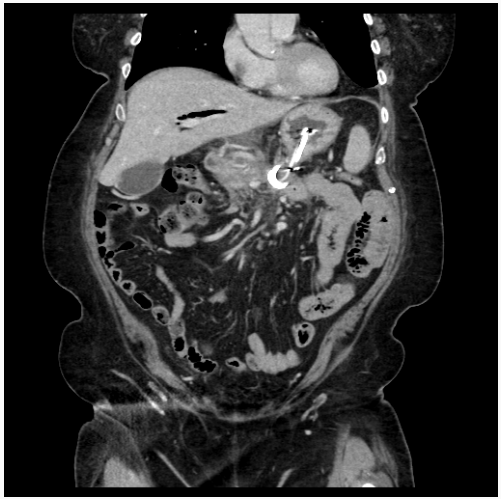


WON—67 y.o. woman with tirzepatide-induced necrotizing pancreatitis





WON—67 y.o. woman with tirzepatide-induced necrotizing pancreatitis



Disconnected pancreatic duct syndrome

Overview

- 28–74% of severe necrotizing pancreatitis
- Lead to recurrent fluid collections, recurrent pancreatitis, pancreatic fistulas

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Verma et al. Pancreatology. 2020;20(6):1035–1044.
Basha et al. Gastrointest Endosc Clin N Am. 2023;33(4):753–770.
Rana et al. Gastroenterol Clin North Am. 2025;54(1):75–95.
Wilcox et al. Clin Gastroenterol Hepatol. 2025;23(11):1878–1887.e3.

Disconnected pancreatic duct syndrome

Overview

- 28–74% of severe necrotizing pancreatitis
- Lead to recurrent fluid collections, recurrent pancreatitis, pancreatic fistulas

Diagnosis

- CT/MRCP: pancreatic necrosis $\geq$2 cm with viable pancreas upstream
- ERCP: disrupted PD with leakage of contrast in the early phases
- EUS: PD terminating into PFC; abrupt cut-off of dilated PD in upstream pancreas

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Verma et al. Pancreatology. 2020;20(6):1035–1044.
Basha et al. Gastrointest Endosc Clin N Am. 2023;33(4):753–770.
Rana et al. Gastroenterol Clin North Am. 2025;54(1):75–95.
Wilcox et al. Clin Gastroenterol Hepatol. 2025;23(11):1878–1887.e3.

Disconnected pancreatic duct syndrome

Management

- Surgery: left pancreatectomy vs. internal drainage (cystojejunostomy)
- EUS-guided pseudocyst/WON drainage $\rightarrow$ Long-term double pigtail plastic stent
- ERCP not recommended: bridging stent successful in only 0–55%

Verma et al. Pancreatology. 2020;20(6):1035–1044.
Maatman et al. J Surg Res. 2020;247:297–303.
Arvanitakis et al. Endoscopy. 2018;50(5):524–546.

Chronic Pancreatitis

Chronic pancreatitis

Definition

- Pathologic fibroinflammatory syndrome of the pancreas in individuals with genetic, environmental and/or other risk factors who develop persistent pathologic responses to parenchymal injury or stress.
- Common features include pancreatic atrophy, fibrosis, pain, duct distortion/strictures, calcifications, exocrine pancreatic insufficiency, diabetes mellitus, and dysplasia.

Whitcomb et al. Pancreatology. 2016;16(2):218–24.

Chronic pancreatitis

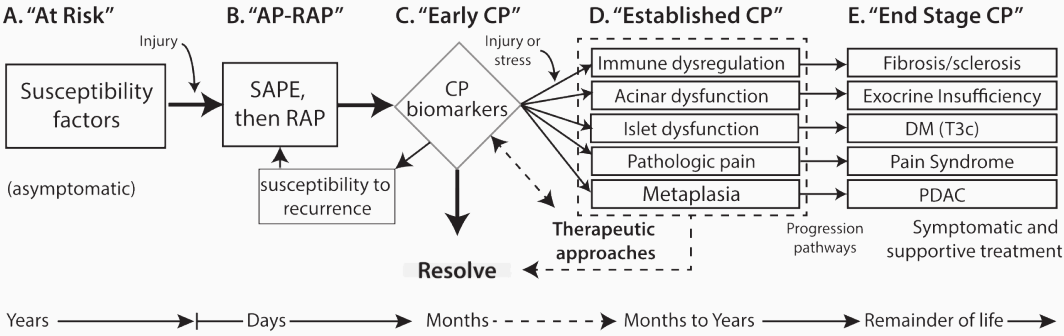
Etiology/Contributors (TIGAR-O)

- Toxic-metabolic: alcohol, tobacco, hypercalcemia, hypertriglyceridemia
- Idiopathic: early-onset, late-onset
- Genetic: *PRSS1*, *SPINK1*, *CFTR*, *CASR*, *CPA1*, *CTRC*, *TRPV6*
- Autoimmune: AIP type 1 (IgG4-related disease), AIP type 2
- Recurrent acute pancreatitis or severe acute pancreatitis
- Obstructive: pancreas divisum, ampullary stenosis, strictures, tumors, IPMNs

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Whitcomb et al. Clin Transl Gastroenterol. 2019;10(6):e00027.

Chronic pancreatitis—Conceptual model



AP, acute pancreatitis; RAP recurrent acute pancreatitis; CP, chronic pancreatitis; DM (T3c), diabetes mellitus type IIIc; PDAC, pancreatic ductal adenocarcinoma; SAPE sentinel acute pancreatitis event.

Whitcomb et al. Pancreatology. 2016;16(2):218–24.

Diagnosis

Tests of pancreatic structure

- CT abdomen
- MRI/MRCP $\pm$ secretin
- EUS
- ERCP

Tests of pancreatic function

- Fecal elastase-1
- 48-hour fecal fat
- Secretin stimulation test

Gardner et al. Am J Gastroenterol. 2020;115(3):322–339.
Conwell et al. Pancreas. 2014;43(8):1143–62.

Pain management

Pathophysiology

- Increased ductal/parenchymal pressure, ischemia, and inflammation
- Altered peripheral and central nociceptive nerve dysfunction

Management

- Acetaminophen, NSAIDs, neuromodulators, opiates
- Endoscopy: EUS with celiac plexus block, ERCP (PD stones, strictures)
- Extracorporeal shock wave lithotripsy (ESWL)
- Surgery: Whipple, modified Puestow (Partington-Rochelle)

Drewes et al. Pancreatology. 2017;17(5):720–731.
Tenner et al. Am J Gastroenterol. 2024;119(3):419–437.

Endoscopic decompression vs. surgery

Cahen et al.

- 39 patients randomized to ERCP vs. pancreaticojejunostomy
- At 2 years:
  - Surgery provided better pain relief (75% vs. 32%; $p = 0.007$)
- At 5 years:
  - Pain relief still better in surgical group (80% vs 38%; $p = 0.042$)
  - 68% in endoscopy group required additional drainage (vs. 5% in surgical group)
  - 47% in endoscopy group eventually underwent surgery

Cahen et al. N Engl J Med. 2007;356(7):676–84.
Cahen et al. Gastroenterology. 2011;141(5):1690–5.

Endoscopic decompression vs. surgery

Issa et al. (ESCAPE trial)

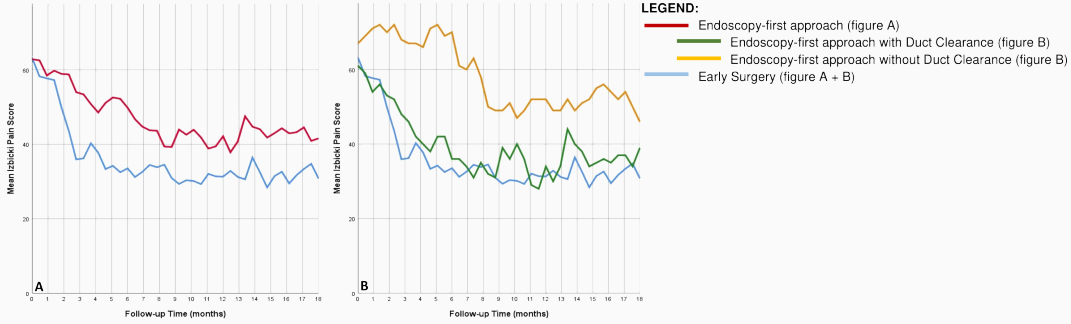
- 88 patients randomized to ERCP vs. pancreaticojejunostomy
- At 18 months: early surgery group had lower Izbicki pain scores (37 vs. 49, P = 0.02)

Issa et al. JAMA. 2020;323(3):237–247.

Endoscopic decompression vs. surgery

Issa et al. (ESCAPE trial)

- 88 patients randomized to ERCP vs. pancreaticojejunostomy
- At 18 months: early surgery group had lower Izbicki pain scores (37 vs. 49, P = 0.02)
- ERCP group achieving duct clearance did as well as surgical group



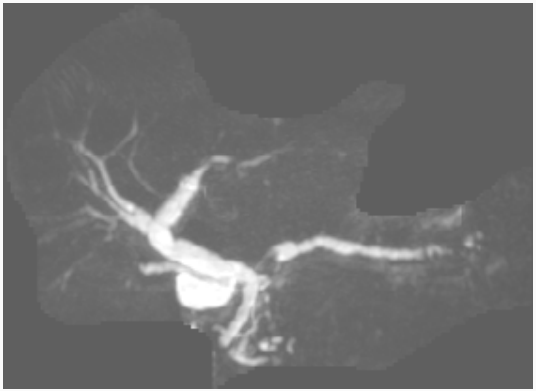
Issa et al. JAMA. 2020;323(3):237–247.

Approach to painful PD stones/strictures

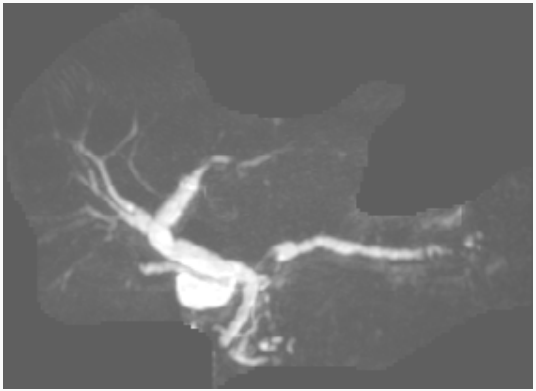
Pancreatic head/neck stones	
Stones $<$5 mm	ERCP
Radiopaque stones $\geq$5 mm	ESWL $\pm$ ERCP *or* ERCP with EHL
Radiolucent stones $\geq$5 mm	ERCP with EHL
Pancreatic tail stones	
All stones	Surgery preferred
Pancreatic duct strictures	
Dominant strictures (head/body)	ERCP with stenting
Dominant strictures (tail)	Surgery
Refractory dominant strictures	Surgery

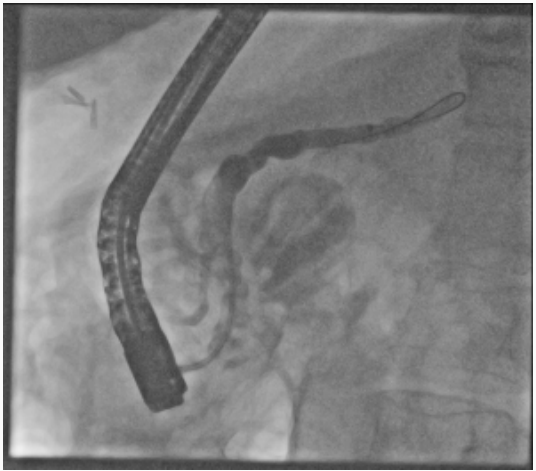
Sheth et al. Gastrointest Endosc. 2024;100(4):584–594.
Dumonceau et al. Endoscopy. 2019;51(2):179–193.
Lohr et al. United European Gastroenterol J. 2017;5(2):153–199.

PD stones—54 y.o. man with alcohol-related chronic pancreatitis



PD stones—54 y.o. man with alcohol-related chronic pancreatitis



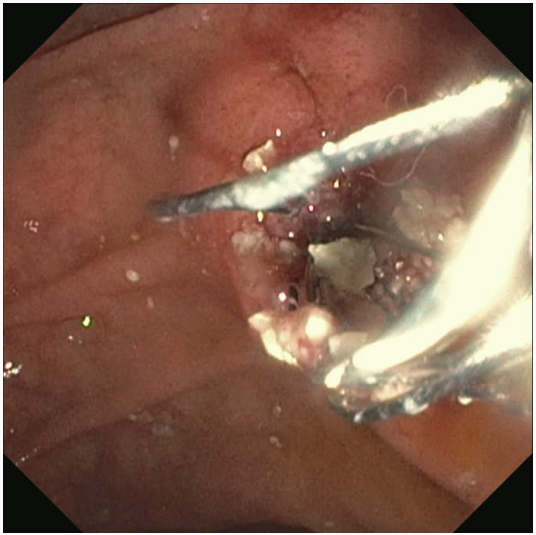


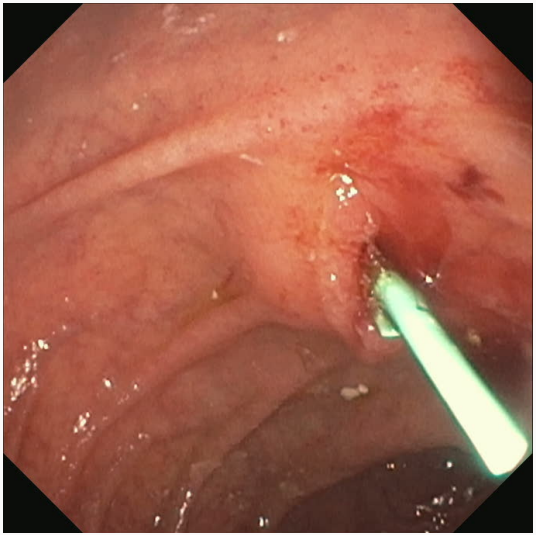
PD stones—54 y.o. man with alcohol-related chronic pancreatitis





PD stones—54 y.o. man with alcohol-related chronic pancreatitis





PD stones—54 y.o. man with alcohol-related chronic pancreatitis



Biliary strictures

Indications for ERCP

- Jaundice
- Persistent cholestasis: alkaline phosphatase $>$2–3 × ULN for $\geq$4 weeks

Technique

- Multiple plastic stents vs. fully-covered self-expanding metal stents (FC-SEMS)
- Stricture resolution achieved in (75–90%) after 6–12 months
- Use of FC-SEMS results in less ERCPs
- May consider multiple plastic stents if low cystic duct take-off with gallbladder in situ

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Sheth et al. Gastrointest Endosc. 2024;100(4):584–594.
Dumonceau et al. Endoscopy. 2019;51(2):179–193.
Lohr et al. United European Gastroenterol J. 2017;5(2):153–199.

Celiac plexus block/neurolysis

Technique

- Region of the celiac plexus injected with an anesthetic and alcohol or a steroid
- EUS-guided vs. percutaneous
  - Better pain relief in EUS group at 4 weeks (81% vs. 35%) and 12 weeks (35% vs. 11%)
- Celiac plexus block (anesthetic + steroid) vs. neurolysis (anesthetic + alcohol)
  - Celiac plexus block generally preferred in benign disease

Durability

- Pain relief 52% at 4 weeks, 57% at 8 weeks, 19% at 12 weeks, 3% at 26 weeks

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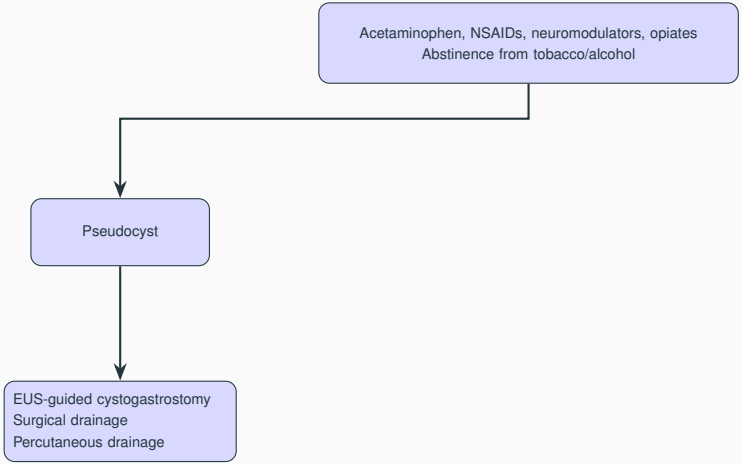
Sheth et al. Gastrointest Endosc. 2025;101(1):e1–e53.
Machicado et al. Pancreatology. 2025;25(6):860–867.

Approach to pain in chronic pancreatitis

Acetaminophen, NSAIDs, neuromodulators, opiates
Abstinence from tobacco/alcohol

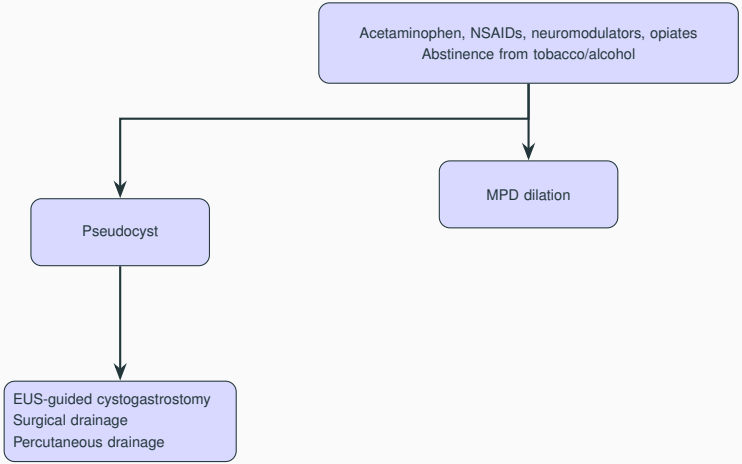
CPB, celiac plexus block; ERCP, endoscopic retrograde cholangiopancreatography; ESWL, extracorporeal shock wave lithotripsy; EUS, endoscopic ultrasound; NSAIDs, non-steroidal anti-inflammatory drugs; MPD, main pancreatic duct; TPIAT, total pancreatectomy with islet autotransplantation.

Approach to pain in chronic pancreatitis



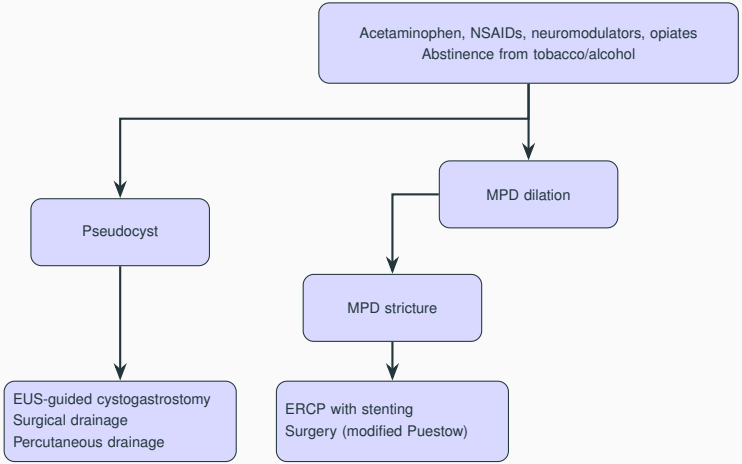
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Approach to pain in chronic pancreatitis



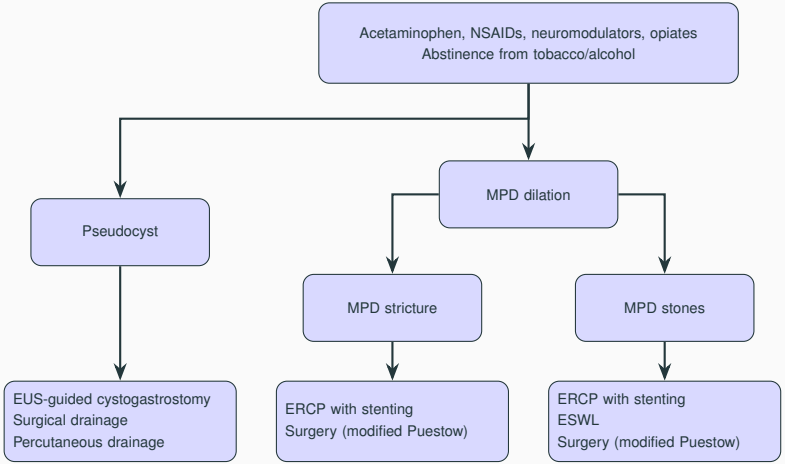
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Approach to pain in chronic pancreatitis



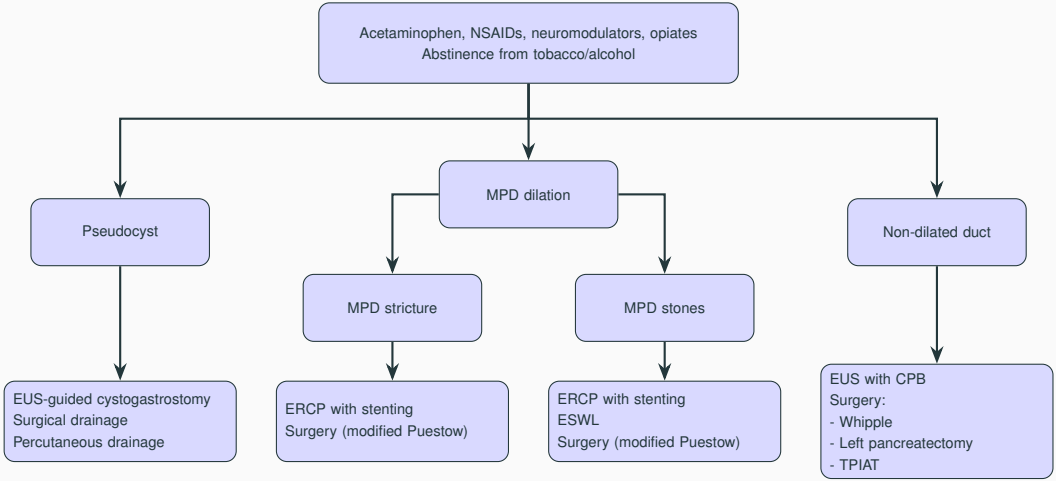
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Approach to pain in chronic pancreatitis



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Approach to pain in chronic pancreatitis



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